



Building and Flashing Code on Windows

README

Windows — Standard Install

1. Download the Windows SDK installer (.exe) from: <https://github.com/zephyrproject-rtos/sdk-ng/releases>
2. Run the installer. The default path usually looks like:

```
C:\zephyr-sdk-0.17.4
```
3. Set environment variables:

```
setx ZEPHYR_TOOLCHAIN_VARIANT zephyr
setx ZEPHYR_SDK_INSTALL_DIR "C:\zephyr-sdk-0.17.4"
```
4. Close PowerShell and open a **new** PowerShell window so the variables take effect.

3. Everyday Usage (After the First Setup)

Whenever you open a new terminal, activate the environment first.

Linux / macOS

```
cd zephyr-esp32s3-course
source .venv/bin/activate
```

Windows (PowerShell)

```
cd zephyr-esp32s3-course
.\.venv\Scripts\Activate.ps1
```

You can now build any course application:

```
python scripts/build.py --app welcome --clean --flash
```

We can also build with the '--monitor' option and this will allow us connect to the board after it reboots with the newly flashed software.

```
python scripts/build.py --app welcome --flash --monitor --port /dev/ttyACM0
```

To exit the serial monitor use the ESP-IDF monitor hotkey: `press Ctrl+]` . You can also try the menu prefix `Ctrl+T` then `Ctrl+X` to quit.

To build in PowerShell we can use:

Key idea: West provides repeatable build/flash/debug commands.